

Ava Farkash

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Education

Cornell University (Ithaca, NY)

Expected May 2027

Bachelors of Science - Mechanical Engineering

Relevant Courses: Statics and Mechanics of Solids, Dynamics, Introduction to Mechanical Design, Intro Fluid Mechanics, Mechanics of Engineering Materials, System Dynamics, Heat Transfer, Mechatronics

Technical Projects

- Carbon Fiber Fuselage Fabrication - Design Build Fly *January 2025 and 2026*
 - Learned and applied an epoxy based carbon fiber layup method with custom 3D printed molds to produce a custom fuselage
- Passenger Compartment Creation - Design Build Fly *September 2025-February 2026*
 - Designed/CADed (Solidworks), prototyped, and constructed (3D printed ABS) a passenger compartment to ensure security within the fuselage
- Arduino Competition Robot - Mechatronics *April-May 2026*
 - Strategized, designed, constructed, and coded an 8x8 inch Arduino Uno robot in a team of three; learned to work within a small budget and delegate tasks to team members; participated in end of year class competition with over sixty teams
- Deflection Testing - Rigidized Metals Internship *June-August 2026*
 - Sketched, laser cut, and assembled a stainless steel testing rig to collect deflection data on 6x6 inch samples and interpreted improvements relative to pattern/texture in different gauges of metal

Professional Experience

Mechanical Sub-team Member

October 2024-Present

- Responsible for designing and manufacturing the fuselage, landing gear, and passenger/cargo requirements; work in tandem with aerodynamics and propulsion sub-teams to integrate their components; participated in the 2026 AIAA DBF competition in Wichita, Kansas

Arc Building Partners - Field Operations Intern

June - August 2025

- Designed and implemented a comprehensive hard drive system to archive all closeout documentation for the Albright Knox Gundlach Art Museum project; presented its structure and usage guidelines to the Director of Facilities, Planning, and Management
- Shadowed project engineers, head safety officers, and project superintendents at various sites

Rigidized Metals - Engineering Intern

June - August 2026

- Learned to use an optical profilometer to start creating a detailed catalogue of *Rigidized®* textures
- Created the first repeatable method for custom pattern modeling using Blender, Inventor, and AutoCAD
- Built deflection testing rig to test most popular textures; presented findings to Director of Engineering

Campus Involvement

- Volunteer Sitter - Guiding Eyes for the Blind *September 2024-Present*
- Full Technical Director - Cog Dog Theater Troupe *August 2025-Present*
- Safety Officer - Design Build Fly *April 2026-Present*

Specialized Skills

Solidworks, Fusion 360, Inventor, AutoCAD, Ansys, Vectorworks, Procore, Microsoft Office Utilities, Basic Python and MatLab, Red Apron Trained- Emerson Manufacturing Lab, machine shop and hand power-tools